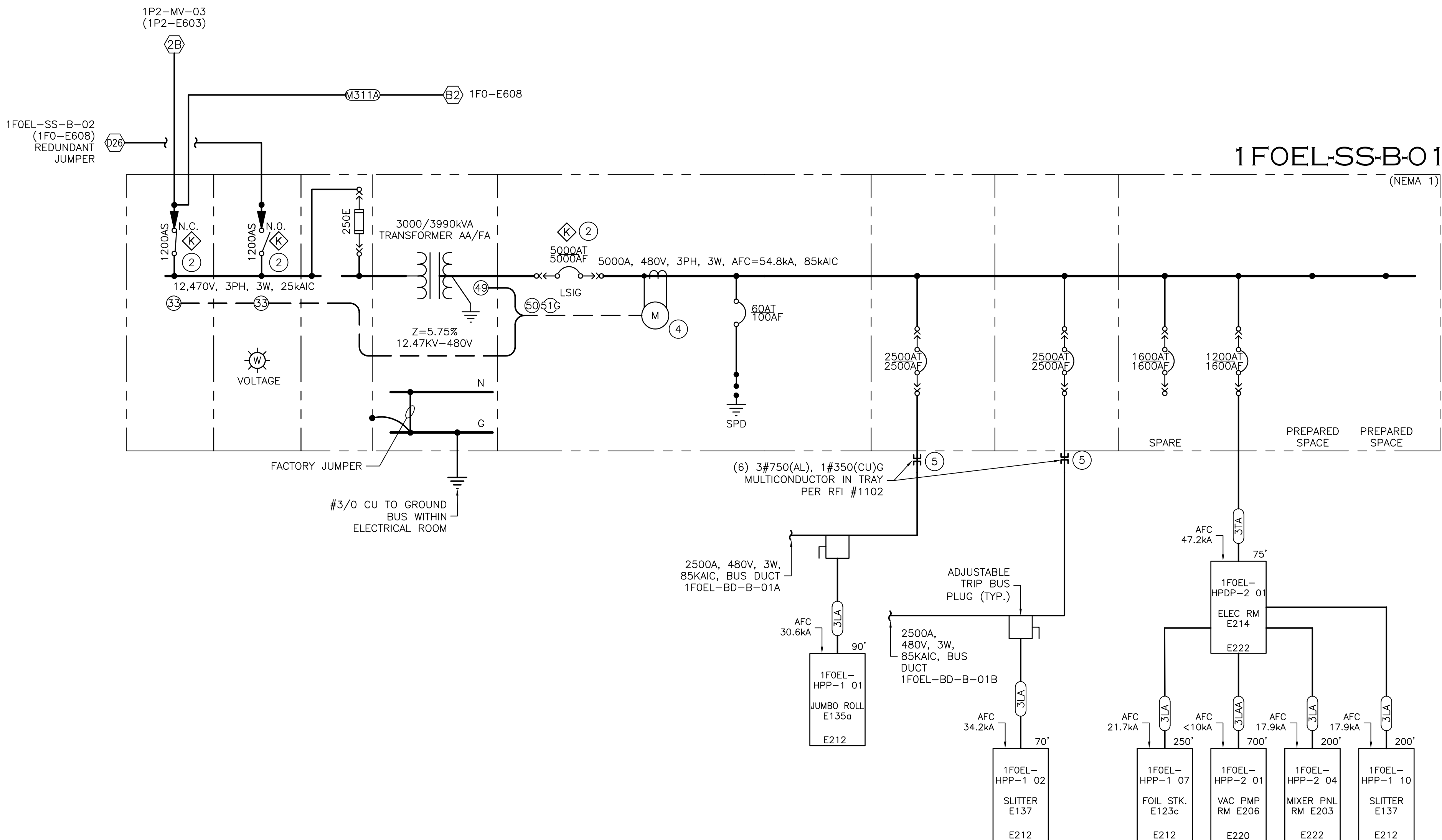




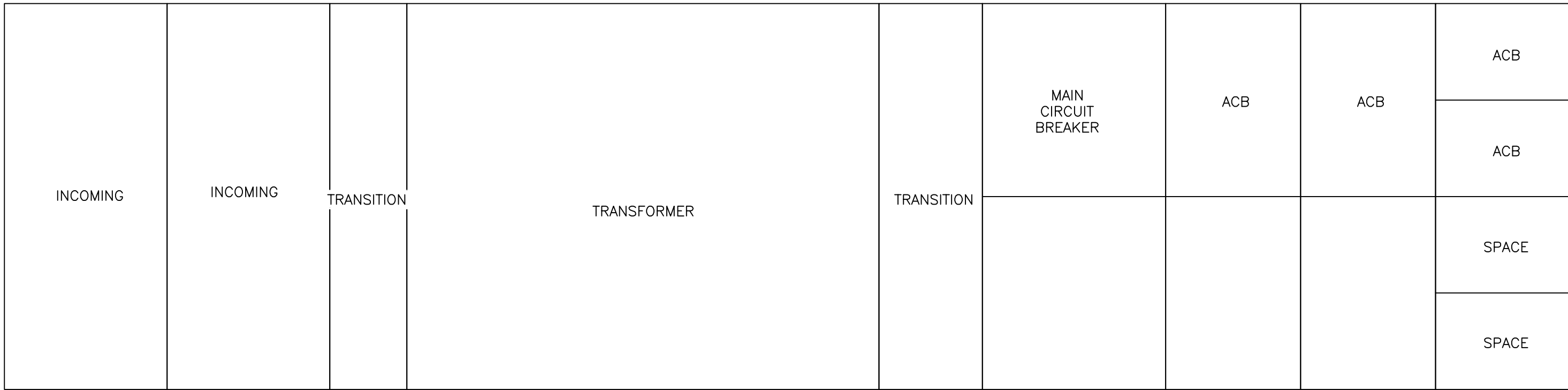
1 ONE LINE DIAGRAM - (1FO) ELECTRODE "B" (PARTIAL)
NO SCALE

BUS DUCT NAME											
1FOEL-BD-B-01A											
		RATINGS				LOADING (KVA)					
BUS PLUG				AMP	AMP			DEM	DEM		
	EQUIPMENT	TAG #	VOLTS	FRAME	TRIP	LOAD	DEM	LOAD	FLA	NOTES	
	INVERTER PANEL	1	480	600	500	377	0.4	151	453		
	CONTROL/DRIVE PANEL	2	480	150	40	22	0.4	9	27		
	INVERTER PANEL	3	480	600	500	377	0.4	151	453		
	CONTROL/DRIVE PANEL	4	480	150	40	22	0.4	9	27		
	INVERTER PANEL	5	480	600	500	377	0.4	151	453		
	CONTROL/DRIVE PANEL	6	480	150	40	22	0.4	9	27		
	INVERTER PANEL	7	480	600	500	377	0.4	151	453		
	CONTROL/DRIVE PANEL	8	480	150	40	22	0.4	9	27	FUTURE	
	INVERTER PANEL	9	480	600	500	377	0.4	151	453	FUTURE	
	CONTROL/DRIVE PANEL	10	480	150	40	22	0.4	9	27	FUTURE	
	INVERTER PANEL	11	480	600	500	377	0.4	151	453	FUTURE	
	CONTROL/DRIVE PANEL	12	480	150	40	22	0.4	9	27	FUTURE	
	INVERTER PANEL	13	480	600	500	377	0.4	151	453	FUTURE	
	CONTROL/DRIVE PANEL	14	480	150	40	22	0.4	9	27	FUTURE	
	INVERTER PANEL	15	480	600	500	377	0.4	151	453	FUTURE	
	CONTROL/DRIVE PANEL	16	480	150	40	22	0.4	9	27	FUTURE	
	1FOEL-HPP-1 01	17	480	400	250	15	1.0	15	18		
	CONTROL/DRIVE PANEL	18	480	150	40	22	0.4	9	27		
	CONTROL/DRIVE PANEL	19	480	150	40	22	0.4	9	27		
	CONTROL/DRIVE PANEL	20	480	150	40	22	0.4	9	27		
	CONTROL/DRIVE PANEL	21	480	150	40	22	0.4	9	27		
	CONTROL/DRIVE PANEL	22	480	150	40	22	0.4	9	27		
	CONTROL/DRIVE PANEL	23	480	150	40	22	0.4	9	27		
	CONTROL/DRIVE PANEL	24	480	150	40	22	0.4	9	27		
	ANODE(-) MAIN PANEL CHILLER #12	25	480	250	250	182	0.4	73	219		
	ANODE(-) R2R DRYER MAIN PANEL #12	26	480	400	400	253	0.4	101	304		
NOTES:											
*800A AND ABOVE BUS PLUGS REQUIRE MANUFACTURER INSTALLED BUS PLATE.											
SEE POWER PLAN FOR LOCATION											
TOTAL LOAD (KVA)						3800	1529		TOTAL DEMAND LOAD (KVA)		
							1839.04		DEMAND AMPS		

3 BUS DUCT LOAD SCHEDULE
NO SCALE

BUS DUCT NAME											
1FOEL-BD-B-01B											
		RATINGS				LOADING (KVA)					
BUS PLUG				AMP	AMP			DEM	DEM		
EQUIPMENT		TAG #	VOLTS	FRAME	TRIP	LOAD	DEM	LOAD	FLA	NOTES	
ANODE(-) MAIN PANEL CHILLER #1		1	480	250	250	182	0.4	73	219		
ANODE(-) R2R DRYER MAIN PANEL #1		2	480	400	400	253	0.4	101	304		
ANODE(-) MAIN PANEL CHILLER #2		3	480	250	250	182	0.4	73	219		
ANODE(-) R2R DRYER MAIN PANEL #2		4	480	400	400	253	0.4	101	304		
ANODE(-) MAIN PANEL CHILLER #3		5	480	250	250	182	0.4	73	219		
ANODE(-) R2R DRYER MAIN PANEL #3		6	480	400	400	253	0.4	101	304		
ANODE(-) MAIN PANEL CHILLER #4		7	480	250	250	182	0.4	73	219		
ANODE(-) R2R DRYER MAIN PANEL #4		8	480	400	400	253	0.4	101	304		
ANODE(-) MAIN PANEL CHILLER #5		9	480	250	250	182	0.4	73	219		
ANODE(-) R2R DRYER MAIN PANEL #5		10	480	400	400	253	0.4	101	304		
ANODE(-) MAIN PANEL CHILLER #6		11	480	250	250	182	0.4	73	219		
ANODE(-) R2R DRYER MAIN PANEL #6		12	480	400	400	253	0.4	101	304		
ANODE(-) MAIN PANEL CHILLER #7		13	480	250	250	182	0.4	73	219		
ANODE(-) R2R DRYER MAIN PANEL #7		14	480	400	400	253	0.4	101	304		
ANODE(-) MAIN PANEL CHILLER #8		15	480	250	250	182	0.4	73	219		
ANODE(-) R2R DRYER MAIN PANEL #8		16	480	400	400	253	0.4	101	304		
ANODE(-) MAIN PANEL CHILLER #9		17	480	250	250	182	0.4	73	219		
ANODE(-) R2R DRYER MAIN PANEL #9		18	480	400	400	253	0.4	101	304		
ANODE(-) MAIN PANEL CHILLER #10		19	480	250	250	182	0.4	73	219		
ANODE(-) R2R DRYER MAIN PANEL #10		20	480	400	400	253	0.4	101	304		
ANODE(-) MAIN PANEL CHILLER #11		21	480	250	250	182	0.4	73	219		
ANODE(-) R2R DRYER MAIN PANEL #11		22	480	400	400	253	0.4	101	304		
1FOEL-HPP-1 02		25	480	400	250	61	1.0	61	74		
NOTES:											
*800A AND ABOVE BUS PLUGS REQUIRE MANUFACTURER INSTALLED BUS PLATE.				TOTAL LOAD (KVA)		4847		1976		TOTAL DEMAND LOAD (KVA)	
SEE POWER PLAN FOR LOCATION								2376.57		DEMAND AMPS	
								2500		RATING (A)	
								95		% LOADING	

2 EQUIPMENT ELEVATION
NO SCALE



GENERAL NOTES

- REFERENCE FEEDER SCHEDULE, E011
- CABLE LENGTHS WHEN INDICATED ON ONE-LINE ARE APPROXIMATE, AND ARE FOR REFERENCE ONLY FOR CALCULATIONS AND ARE NOT TO BE USED FOR MATERIAL TAKEOFFS.
- SECONDARY UNIT SUBSTATIONS: REFER TO SPECIFICATION #26 11 16 12.
- UNIT SUBSTATION SCADA COMMUNICATIONS SHALL INCLUDE ALL HARDWARE DEVICES TO COMMUNICATE I/O (VIA MODBUS TCP COMMUNICATIONS) INCLUDING, BUT NOT LIMITED TO, THE FOLLOWING:
 - PRIMARY SWITCH STATUS (OPEN/CLOSED)
 - TRANSFORMER WINDING TEMPERATURE
 - TRANSFORMER FAN RUN STATUS
 - BREAKER STATUS
 - POWER METER DATA
- ALL SWITCHBOARDS, SWITCHGEAR, AND PANELBOARDS SUPPLIED BY FEEDERS SHALL BE PERMANENTLY MARKED TO INDICATE EACH DEVICE OR EQUIPMENT WHERE THE POWER ORIGINATES. THE LABEL SHALL BE PERMANENTLY AFFIXED, OF SUFFICIENT DURABILITY TO WITHSTAND THE ENVIRONMENT INVOLVED, AND NOT HANDWRITTEN.

KEYED NOTES

- NOT USED
- PRIOR TO CLOSING A SWITCH, THE OTHER SWITCH MUST BE OPENED USING KIRK KEY INTERLOCK TO PREVENT CONNECTION OF MULTIPLE SOURCES. REFER TO SEQUENCE OF OPERATIONS BY MANUFACTURER IN EQUIPMENT SUBMITTAL.
- NOT USED.
- SQUARE D/ SCHNEIDER ION 9000 WITH RS-485 COMMUNICATIONS.
- ENGINEERED CABLE BUS ASSEMBLY, CONDUIT, OR CABLE TRAY TO BUS DUCT CABLE TAP BOX.

PROJECT ALPHA 1F0 - MAIN MANUFACTURING BUILDING EAST PECOS ROAD QUEEN CREEK, ARIZONA 85140

ELECTRICAL ONE LINE DIAGRAM

DATE	REMARKS
10/19/2024	1F0 CD PROGRESS SET (MEP ONLY)
11/01/2024	1F0 CD PROGRESS SET (MEP ONLY)
12/17/2024	1F0 CD PROGRESS SET (MEP ONLY)
03/12/2025	1F0 CD PROGRESS SET (MEP ONLY)
07/24/2025	1F0 CD PROGRESS SET (MEP ONLY)
10/23/2025	1F0 CD PROGRESS SET (MEP ONLY)

P/APP:	WB
DRAWN BY:	ETS/AO/AUG
JOB NO:	PHX21-5038-00

SHEET 1F0-E607